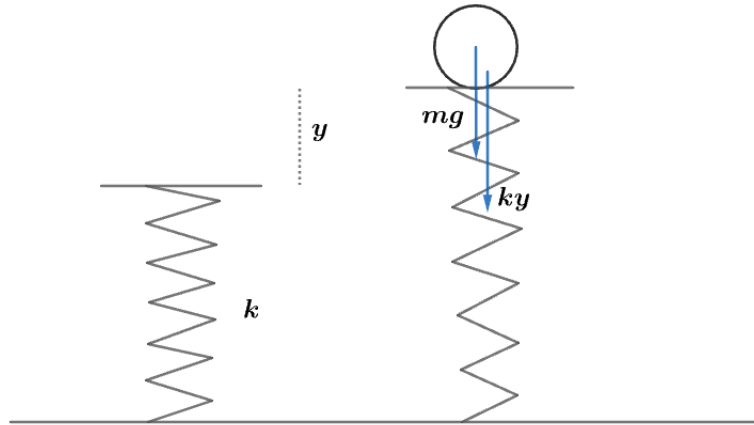


2008 F=ma Exam: Problem 16

Kevin S. Huang



Applying Newton's 2nd law to the ball when it is at height y above the original position of the spring,

$$F = -mg - ky = ma$$

taking downwards to be negative as indicated. We have

$$a = -g - \frac{ky}{m}$$

so the answer is E.